Lösungen · Trigonometrie

1. $\varepsilon = \cos^{-1}(\frac{13}{32}) \approx 66{,}0°$ ..... $\cos\varepsilon = \frac{13}{32}$

2. $\beta_1 = \cos^{-1}(\frac{255}{384}) \approx 48{,}4°$ ..... $\cos\beta_1 = \frac{255}{384}$

3. $\alpha = \sin^{-1}(\frac{7{,}4}{14{,}1}) \approx 31{,}7°$ ..... $\sin\alpha = \frac{7{,}4}{14{,}1}$

4. $\alpha = \cos^{-1}(\frac{4}{9}) \approx 63{,}6°$ ..... $\cos\alpha = \frac{4}{9}$

5. $\alpha = \tan^{-1}(\frac{14}{9}) \approx 57{,}3°$ ..... $\tan\alpha = \frac{14}{9}$

6. $x = 2\sqrt{7289} \approx 170{,}8$ cm; $\alpha = \tan^{-1}(\frac{16}{170}) \approx 5{,}4°$ ...
$x^2 = 170^2 + 16^2$; $\tan\alpha = \frac{16}{170}$

7. $BC = \sqrt{20} \approx 4{,}47$; $\beta = \tan^{-1}(2) \approx 63{,}4°$ ... $BC^2 = 2^2 + 4^2$; $\tan\beta = \frac{4}{2}$

8. $a = \frac{7{,}4}{\sin 52°} \approx 9{,}4$ m ..... $\sin 52° = \frac{7{,}4}{a}$

9. a) **AB ≈ 1228,7 m** ..... $\sin 55° = AB : 1500$

b) **AD ≈ 860,4 m** ..... $\cos 55° = AD : 1500$

10. $PB = \frac{1{,}20}{\tan 34{,}9°} \approx 1{,}72$ m ..... $\tan 34{,}9° = 1{,}20 : PB$

11. $AC = \frac{h}{\sin 22°} \approx 78{,}1$ cm ..... $h = \sqrt{32^2 - 13^2} = 3\sqrt{95}$; $\sin 22° = h : AC$

12. **BF ≈ 93,0 cm** ..... $\sin 45° = BF : 131{,}5$
BC = BF : cos 65° ≈ 220,0 cm ..... $\cos 65° = BF : BC$

13. **Berg = 57,9 m; D = 69,9 m** ..... $\sin 30° = BC : 112{,}8$; Berg = BC + 1,5; D = 57,9 + 12,0

14. **BD ≈ 502 m** ..... $\sin 54° = AD : 920$; $\tan 34° = BD : AD$

15. **100 und 6**
ja ..... $\frac{16}{170} = \frac{8}{85} \approx 9{,}4\ \% > 6\ \%$